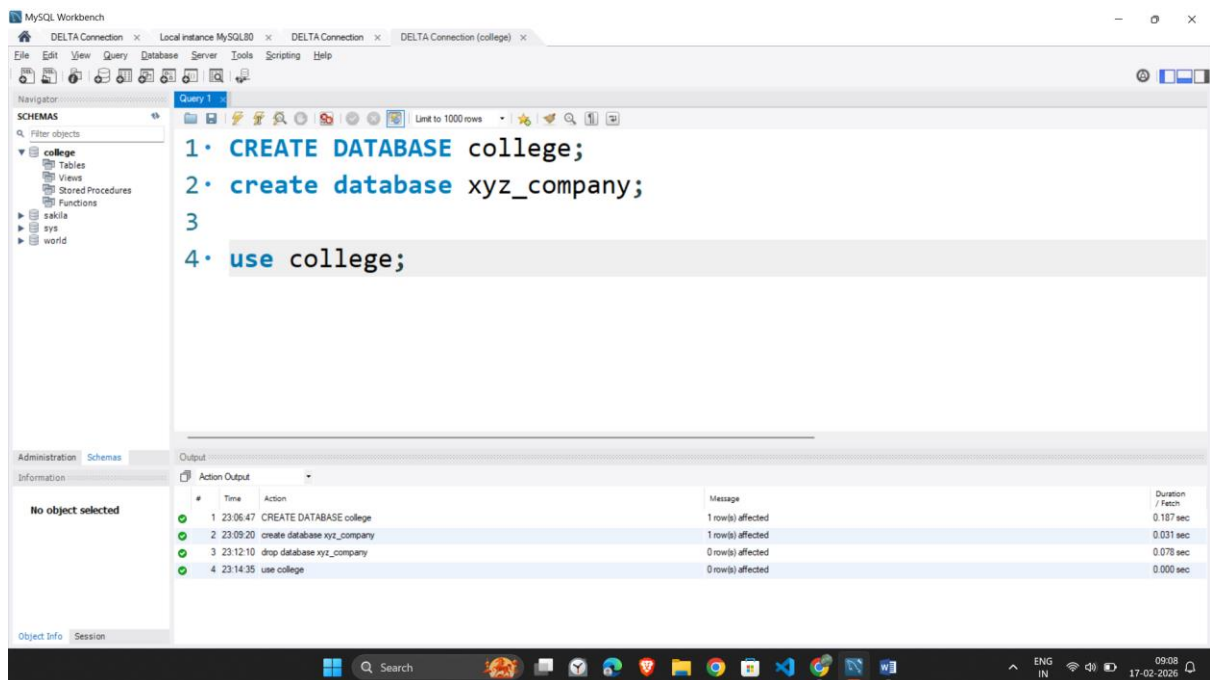
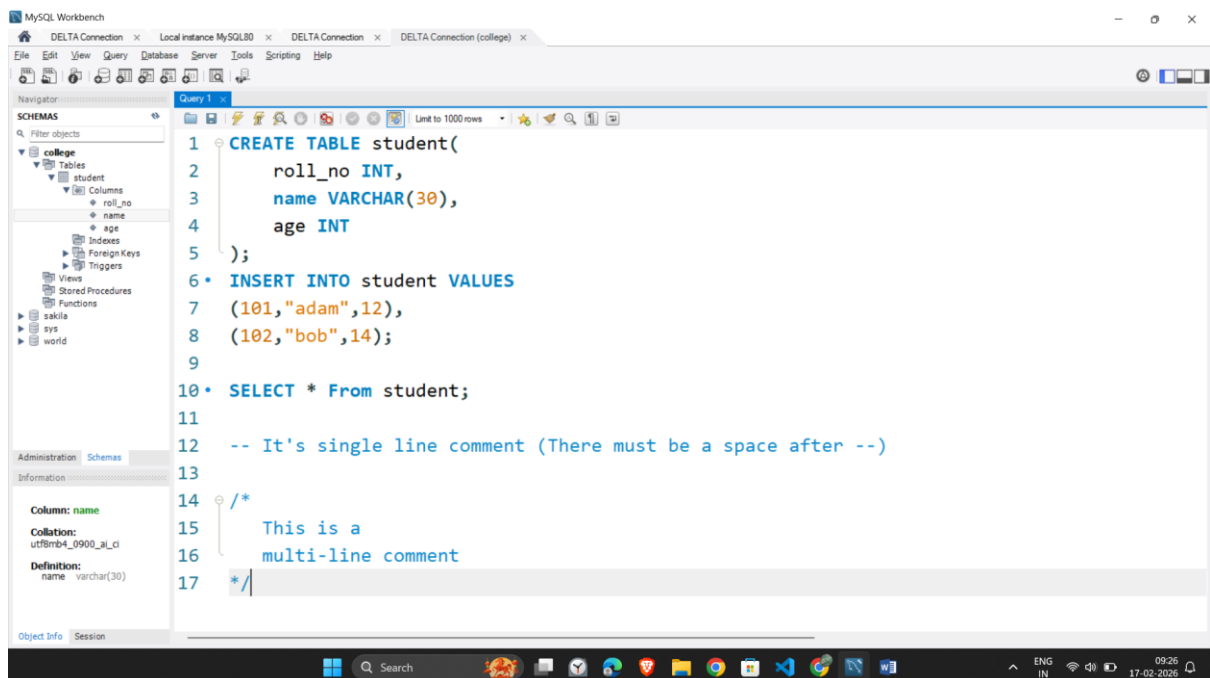


39. Starting with SQL

#TOPIC 7- OUR FIRST DATABASE:



#Topic 8 –Our First Table :



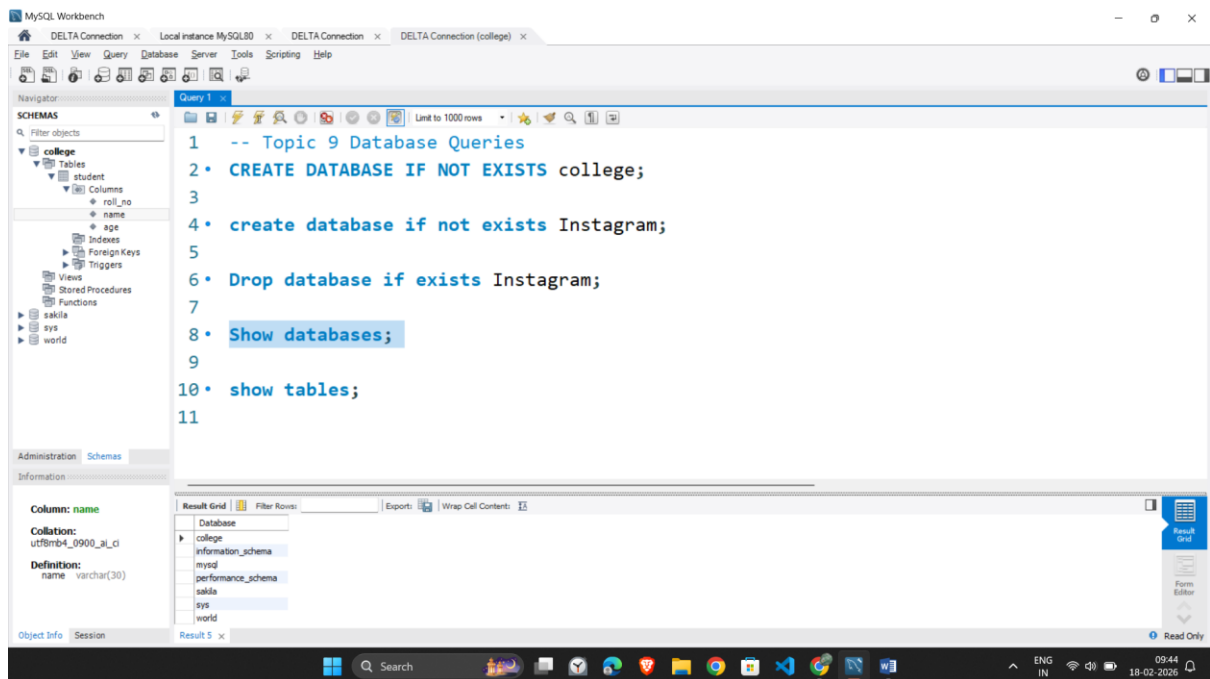
10. SELECT * From student;

11

Result Grid

roll_no	name	age
101	adam	12
102	bob	14

#Topic 9 Database Queries:



```

0 • show tables;
1

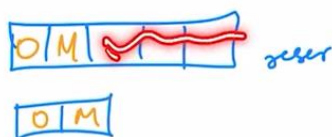
```



#Topic 10 – TABLE QUERIES: Create Table

Data Types

DATATYPE	DESCRIPTION	USAGE
CHAR	string(0-255), can store characters of fixed length	CHAR(50)
VARCHAR	string(0-255), can store characters up to given length	VARCHAR(50)
BLOB	string(0-65535), can store binary large object	BLOB(1000)
INT	integer(-2,147,483,648 to 2,147,483,647)	INT
TINYINT	integer(-128 to 127)	TINYINT
BIGINT	integer(-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807)	BIGINT
BIT	can store x-bit values. x can range from 1 to 64	BIT(2)
FLOAT	Decimal number - with precision to 23 digits	FLOAT
DOUBLE	Decimal number - with 24 to 53 digits	DOUBLE
BOOLEAN	Boolean values 0 or 1	BOOLEAN
DATE	date in format of YYYY-MM-DD ranging from 1000-01-01 to 9999-12-31	DATE
YEAR	year in 4 digits format ranging from 1901 to 2155	YEAR



name CHAR(5)
name VARCHAR(5)

Char = fixed length data type

Varchar= variable length data type i.e can reduce length if requirement is less.

#Topic 11 What are constraints?

Constraints

Rules for data in the table

<u>NOT NULL</u>	columns cannot have a null value
UNIQUE	all values in column are different
DEFAULT	sets the default value of a column
CHECK	it can limit the values allowed in a column

salary **INT DEFAULT 25000**

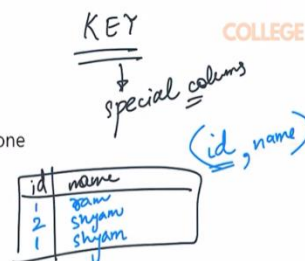
CONSTRAINT age_check **CHECK** (age >= 18 **AND** city="Delhi")

#Topic 12 Key Constraints

Constraints

PRIMARY KEY makes a column unique & not null but used only for one

```
CREATE TABLE temp (  
  id int not null,  
  PRIMARY KEY (id)  
);
```



FOREIGN KEY prevent actions that would destroy links between tables

```
CREATE TABLE temp (  
  cust_id int,  
  FOREIGN KEY (cust_id) REFERENCES customer(id)  
);
```



stu

id	name	bid
101		2
102		1

FK

teach

id	name
1	
2	
3	

PK

FOREIGN KEY (stu_id) REFERENCES teach(id)

User					
id (PK)	age	name	email	followers	following
1		adam	adam@yahoo.in	123	145
2		bob	bob123@gmail.com	200	200
3		casey	casey@email.com	300	306
4		donald	donald@gmail.com	200	105

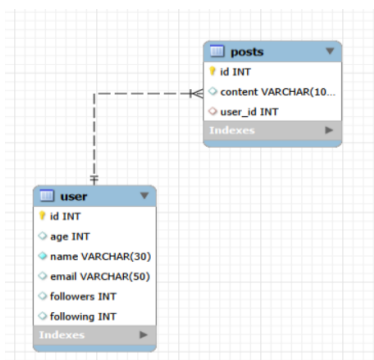
Posts			
	id (PK)	content	user_id (FK)
	101	"Hello World"	3
	102	"Bye Bye"	1

```

3 • USE Instagram ;
4
5 • CREATE TABLE user(
6     id INT,
7     age INT,
8     name VARCHAR(30) NOT NULL,
9     email VARCHAR (50) UNIQUE,
10    followers INT DEFAULT 0,
11    following INT,
12    CONSTRAINT CHECK (age>=18),
13    PRIMARY KEY (id)
14 );

CREATE TABLE posts (
    id INT PRIMARY KEY,
    content VARCHAR(100),
    user_id INT,
    FOREIGN KEY (user_id) REFERENCES user(id)
);

```



ER(Entity relation) Diagram

#TOPIC 13 Primary and Foreign Keys:

What are Keys?

Keys are special columns in the table

Primary Key

It is a column (or set of columns) in a table that uniquely identifies each row. (a unique id)
There is only 1 PK & it should be NOT null.

*unique
not null*

Foreign Key

A foreign key is a column (or set of columns) in a table that refers to the primary key in

FKs can have duplicate & null values.

There can be multiple FKs.

to the primary key in another table.

Primary & Foreign Keys

PK		User			
id	Name	Email	Followers	Following	
1	Adam	adam@yahoo.in	123	145	
2	Bob	bob123@gmail.com	200	200	
3	Casey	casey@email.com	300	306	
4	Donald	donald@gmail.com	200	105	

PK		Posts		FK	
id	Content	User_id			
1	"Hello World"	3			
2	"I am back!"	4			
3	"Bye Bye"	1			

#Topic 14 INSERT into TABLE

INSERT INTO user

(id,age,name,email,followers,following)

VALUES

```
(1,19,"adam","adam@yahoo.in",123,145),  
(2,20,"Bob","bob123@gmail.com",200,200),  
(3,21,"Casey","casey@email.com",300,306),  
(4,22,"Donald","donald@gmail.com",200,105);
```

#Topic 15 SELECT command

31 • **SELECT id,name,email FROM user;**

32

Result Grid				Filter Rows:	Edit:	Export/Import:	Wrap
	id	name	email				
▶	1	adam	adam@yahoo.in				
	2	Bob	bob123@gmail.com				
	3	Casey	casey@email.com				
	4	Donald	donald@gmail.com				
*	NULL	NULL	NULL				

33 • **SELECT * FROM user ;**

34

Result Grid							Filter Rows:	Edit:	Export/Import:	Wrap
	id	age	name	email	followers	following				
▶	1	19	adam	adam@yahoo.in	123	145				
	2	20	Bob	bob123@gmail.com	200	200				
	3	21	Casey	casey@email.com	300	306				
	4	22	Donald	donald@gmail.com	200	105				
*	NULL	NULL	NULL	NULL	NULL	NULL				

35 • **SELECT DISTINCT age from user;**

36

Result Grid		Filter Rows:	Export:	Wrap Cell Contents:
	age			
▶	19			
	20			
	21			
	22			

//DISTINCT WILL GIVE UNIQUE VALUES EXCLUDING DUPLICATE

40.SQL (Part – 2)

#Topic 1 WHERE CLAUSE:-

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Where Clause

Operators

Arithmetic Operators : +(addition) , -(subtraction), *(multiplication), /(division), %(modu

Comparison Operators : = (equal to), != (not equal to), > , >= , < , <=

Logical Operators : AND, OR , NOT, IN, BETWEEN, ALL, LIKE, ANY

Bitwise Operators : & (Bitwise AND), | (Bitwise OR)



x & y	x	y	Result
0 & 0	0	0	0
0 & 1	0	1	0
1 & 0	1	0	0
1 & 1	1	1	1

x | y
↑ pipe

x & y	x	y	Result
0 & 0	0	0	0
0 & 1	0	1	1
1 & 0	1	0	1
1 & 1	1	1	1

x | y
↑ pipe

```
1 use Instagram;
2 select * from user;
3
4 SELECT * FROM user WHERE followers >= 200;
5 SELECT name, followers FROM user Where followers >=200;
6
7 SELECT name, age FROM user Where age<22;
8
9
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
name	age			
adam	19			
Bob	20			
Casey	21			

#Topic 3 Frequently used Operators:=

LOGICAL OPERATOR-

Where Clause

Frequently used Operators

AND (to check for both conditions to be true)

OR (to check for one of the conditions to be true)

BETWEEN (selects for a given range)

IN (matches any value in the list)

NOT (to negate the given condition)

-- WHERE CLAUSE

- `SELECT name,age FROM user WHERE age + 1 = 22; -- right now age is 21`
-- AND
 - `SELECT name,age,followers FROM user WHERE age>=19 AND followers>200;`
-- OR
 - `SELECT name,age,followers FROM user WHERE age>=19 OR followers>200;`
-- BETWEEN
 - `select name,age,followers FROM user WHERE age BETWEEN 19 AND 21;`
-- IN
 - `SELECT name,email FROM user WHERE email IN ("donald@gmail.com","adam@yahoo.in","a@.i");`
-- NOT
- IN
- `SELECT name,email FROM user WHERE email IN ("donald@gmail.com","ad`
 - `INSERT INTO user (id,age,name,email,followers,following) VALUES`
`(5,19,"eve","eve@yahoo.in",400,145),`
`(6,20,"farah","farah@gmail.com",10000,1000);`
 - `SELECT name,followers,age FROM user WHERE age in (19,20);`

Grid	Filter Rows	Export	Wrap Cell Contents
ame	followers	age	
len	123	19	
o	200	20	
e	400	19	
rah	10000	20	

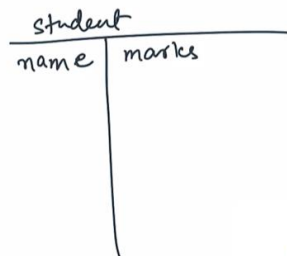
***FROM HERE FOR SQL COMMANDS =SEE
sql files in folder .***

#TOPIC 6 – AGGREGATE FUNCTIONS

Aggregate Functions

Aggregate functions perform a calculation on a set of values, and return a single value.

- COUNT()
- MAX()
- MIN()
- SUM()
- AVG()



student	
name	marks

Example :

```
SELECT max(marks)
FROM student;
```

#TOPIC 7 GROUP BY CLAUSE

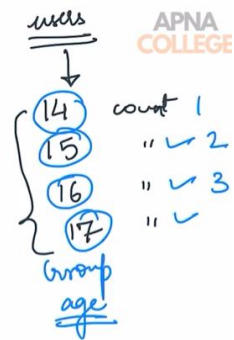
Group by Clause

Groups rows that have the same values into summary rows.

It collects data from multiple records and groups the result by one or more column.

```
SELECT col1, col2
FROM table_name
GROUP BY col_name(s);
```

*Generally we use group by with some *aggregation function*.



#TOPIC 8 HAVING CLAUSE

Having Clause

Similar to Where i.e. applies some condition on rows.

But it is used when we want to apply any **condition after grouping**.

where
(condition)

```
SELECT col1, col2
FROM table_name
GROUP BY col_name(s)
HAVING condition;
```

}

select age, max(followers)
From user
Group By age
Having max(followers) > 200;

babbar

- WHERE is for the table, HAVING is for a group
- Grouping is necessary for HAVING

#TOPIC 9 GENERAL ORDER

General Order

```
SELECT column(s)
FROM table_name
WHERE condition
GROUP BY column(s)
HAVING condition
ORDER BY column(s) ASC;
```

#TOPIC 10 UPDATE TABLE

Table Queries

Update (to update existing rows)

```
UPDATE table_name
SET col1 = val1, col2 = val2
WHERE condition;
```

TOPIC 11 DELETE TABLE

Table Queries

Delete (to delete existing rows)

DELETE FROM *table_name*
WHERE *condition;*

Geeks

#TOPIC 12 ALTER TABLE

Table Queries

Alter (to change the schema)

ADD Column

ALTER TABLE *table_name*
ADD COLUMN *column_name datatype constraint;*

DROP Column

ALTER TABLE *table_name*
DROP COLUMN *column_name;*

RENAME Table

ALTER TABLE *table_name*
RENAME TO *new table name;*

Table Queries

Alter (to change the schema)

CHANGE Column (rename)

ALTER TABLE *table_name*
CHANGE COLUMN *old_name new_name new_datatype new_constraint;*

MODIFY Column (modify datatype/ constraint)

ALTER TABLE *table_name*
MODIFY *col_name new_datatype new_constraint;*



#TOPIC 13 TRUNCATE TABLE

Table Queries

Truncate (to delete table's data)

TRUNCATE TABLE *table_name* ;

Drop table
↓
X

DROP-IT DELETES WHOLE TABLE

DELETE – TO DELETE EXISTING ROWS

#TOPIC 14 – PRACTICE QS 1

Practice Qs

Qs: Create a database for your college.

Create a table named Teacher to store (id, name, subject, salary)

Insert following data in the table :

23, "ajay", "math", 50,000

47, "bharat", "english", 60,000

18, "chetan", "chemistry", 45,000

9, "divya", "physics", 75,000

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Practice Qs

✓ Select teachers whose salary is more than 55K

- Rename the salary column of teacher table to ctc
- Update salary of all teachers by giving them an increment of 25%
- Add a new column for teachers called city. The default city should be "Gurgaon"
- Delete the salary column for teacher table

#TOPIC 15 PRACTICE QS 2

Practice Qs

Qs: Create a table to store student info (roll_no, name, city, marks).

Insert following data in the table :

110, "adam", "Delhi", 76

108, "bob", "Mumbai", 65

124, "casey", "Pune", 94

112, "duke", "Pune", 80

Practice Qs

- Select all students who scored 75+
- Find names of all cities where students are from
- Find the maximum marks of students from each city
- Find the average of the class
- Add a new column grade, assign grade such that :

marks > 80, grade = O

marks 70-80, grade = A

marks 60-70, grade = B
